Recipe Agent

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Roles

Scrum Master - Jalen

Progress Reporting - Will be rotating per week

UI/Coding: Jalen, Pranav, Sarosh

Github Link:

https://github.com/saroshawan100/Recipe-Agent

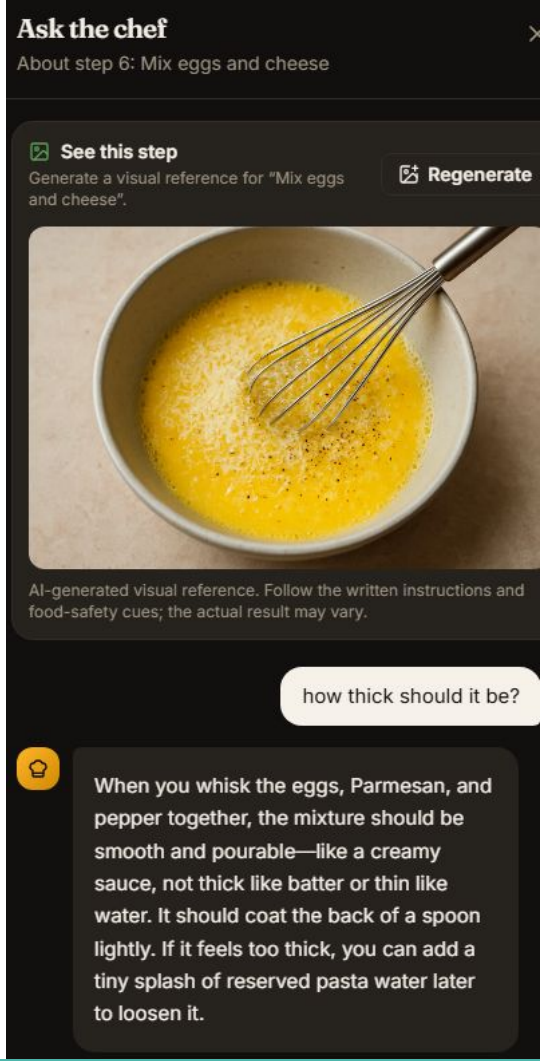
Progress Update

What is New?

Cooking Generation

Output an image for the step when requested to save tokens

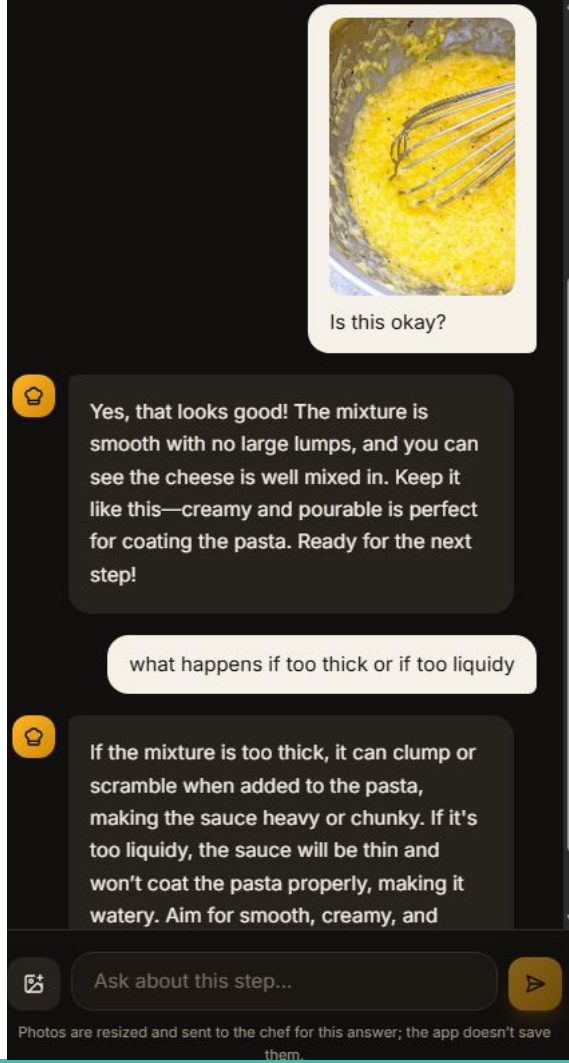
- Clears confusion by showing a target appearance
- Lets users compares to the food being made
- Adds visual clarity



Cooking Recognition

Input images when users are lost and confused

- Recognizes what the food should look like in the current step
- Offers timely feedback to salvage and remedy the situation
- Compare visible cues to recipe’s instructions
- Help reassure users and provides confidence

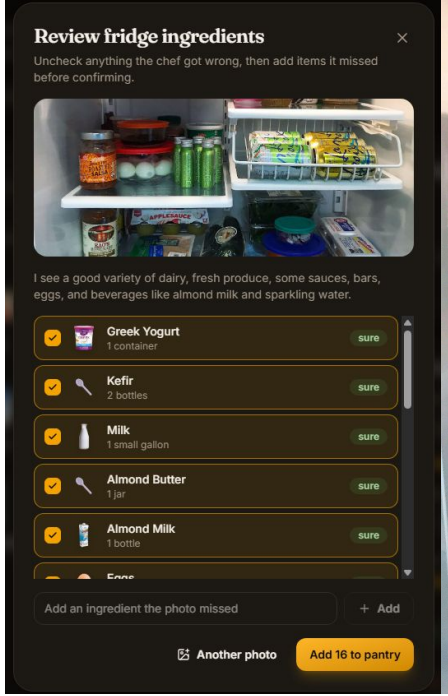


Refrigerator Detection

Scan or upload image for ingredient detection

VLM may be less reliable with unfamiliar regional ingredients, unusual packaging, partly hidden, or look similar to common foods.

User should double check and add or remove ingredients





Gives likelihoods:

Sure — high confidence

Likely — medium confidence

Unsure — low confidence

Evaluation Framework & Key Findings

- Built an evaluation suite for the AI agent: 41 test cases across all six stages (recipe suggestions, full recipes, live coaching, check-ins, in-cook Q&A, chat). Each case is graded by automated rule checks for allergens, diets, time limits, safe cooking temperatures and units, plus an optional AI judge that scores answer quality against a rubric.

- Measured a baseline and found real safety issues: 29 of 41 cases passed (71%). The agent suggested peanut dishes to a user with a peanut allergy, used regular soy sauce in "gluten-free" recipes, ignored time limits on "quick shop" recipes, and left long gaps in live coaching.

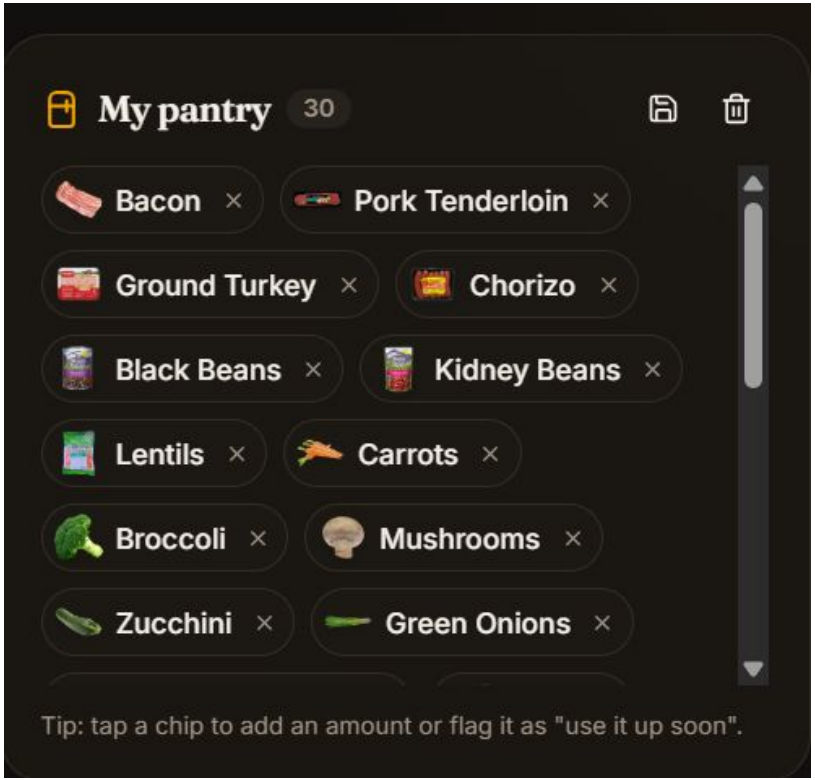
Accounts and Syncing

- **Personalized user accounts:** Added email and password signup and login, with secure password hashing and persistent sessions, so users can return to their own account.
- **Synced pantry and preferences:** Connected each account to its own pantry and cooking preferences, including allergies. The app restores that personal data after login and syncs changes as users update it.
- **MongoDB Atlas:** Added MongoDB storage for account, pantry, and preference data, configured through a private .env connection string. Verified that the backend connects to the Atlas cluster.

Accounts and Syncing Example



- Example of creating an account and preferences being saved to a specific user.



Proposed Next Steps

- **Decide on the recipe bank:** keep it in scope, or rely on the Al model with structured outputs
- **Deploy:** host the app so the whole team and testers can use it without running it locally

Questions?

Thank you!